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(54) **CONNECTOR ASSEMBLY WITH HIGHER SPACE UTILIZATION**

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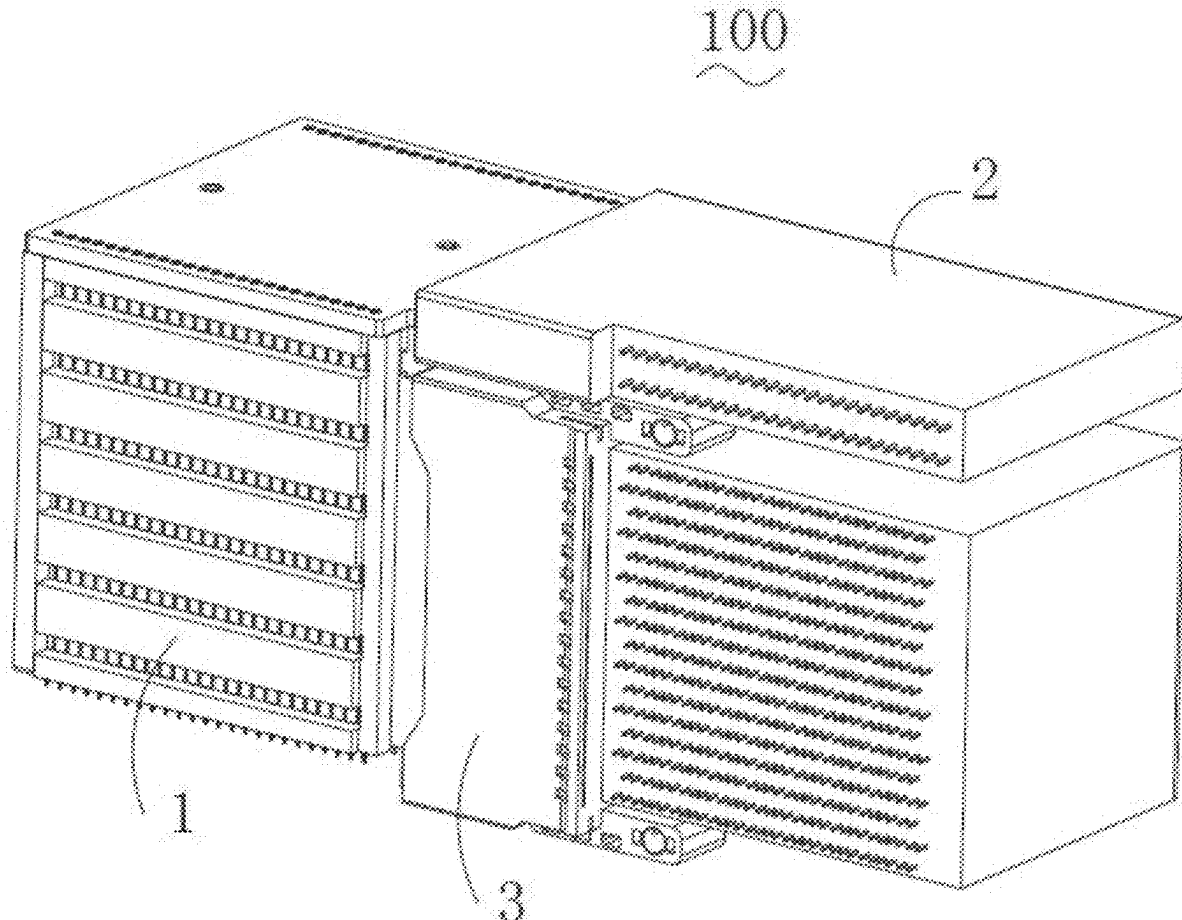
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CPC **H01R 12/721** (2013.01); **H01R 12/714** (2013.01); **H01R 12/732** (2013.01)

(57)

ABSTRACT

A connector assembly includes a card connector and a first backplane connector. The card connector includes a first terminal module, a second terminal module and a circuit board card. The circuit board card connects the first terminal module and the second terminal module to form a receiving cavity. The first backplane connector is received in the receiving cavity. As a result, space utilization of the connector assembly is improved.



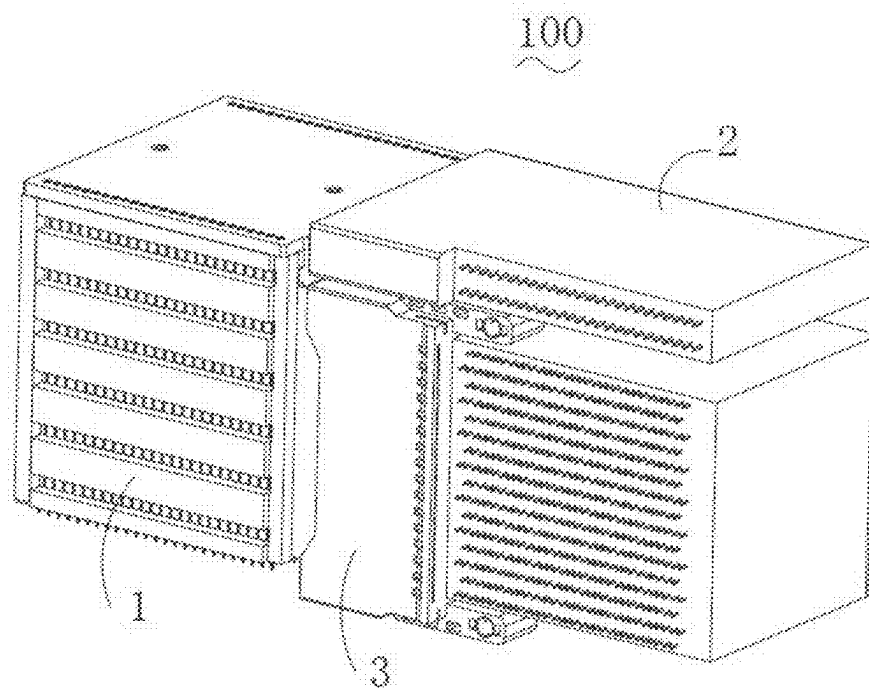


FIG. 1

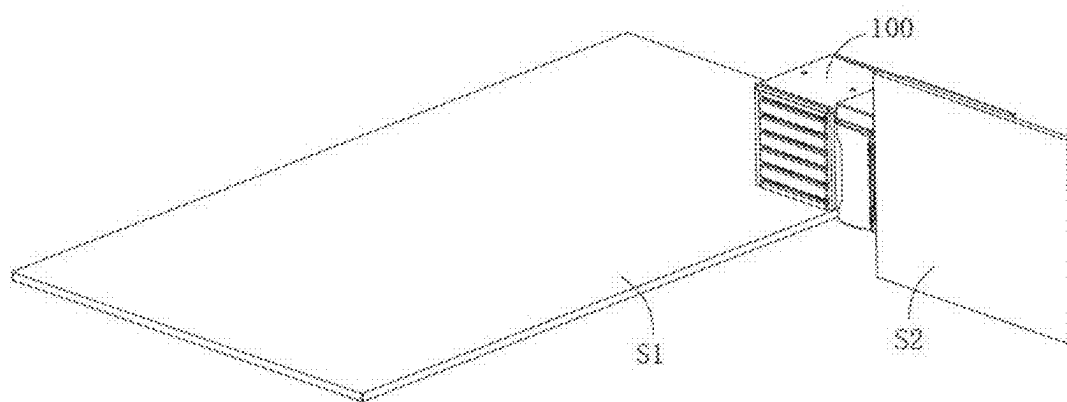


FIG. 2

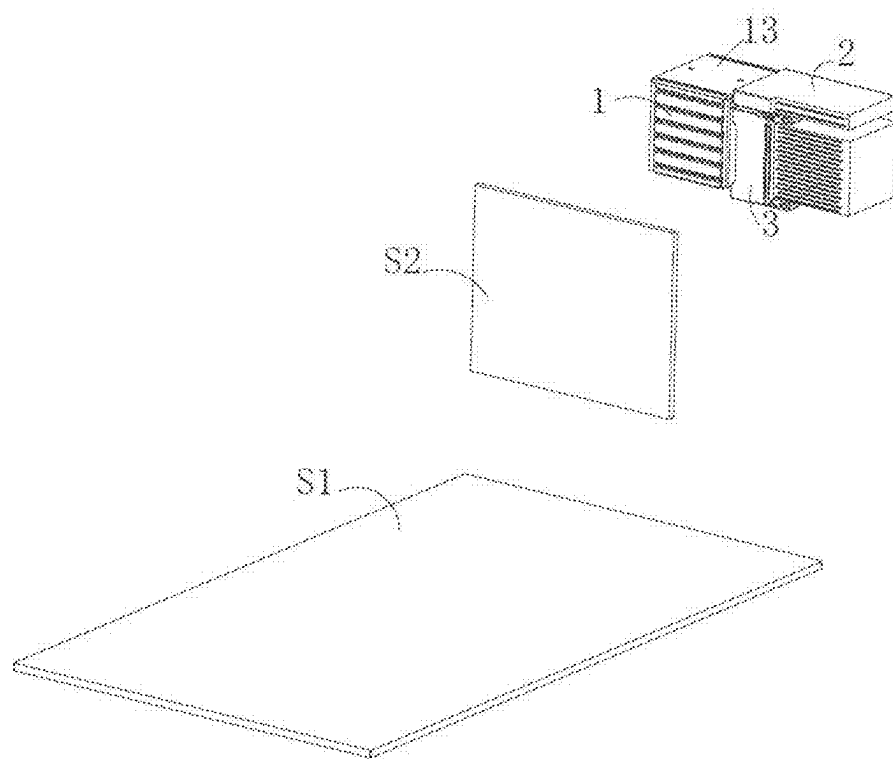


FIG. 3

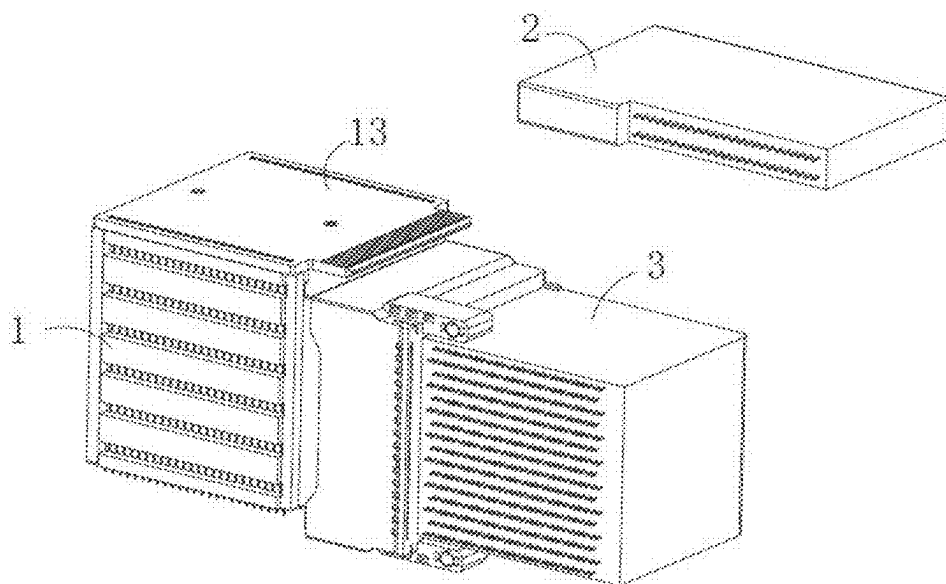


FIG. 4

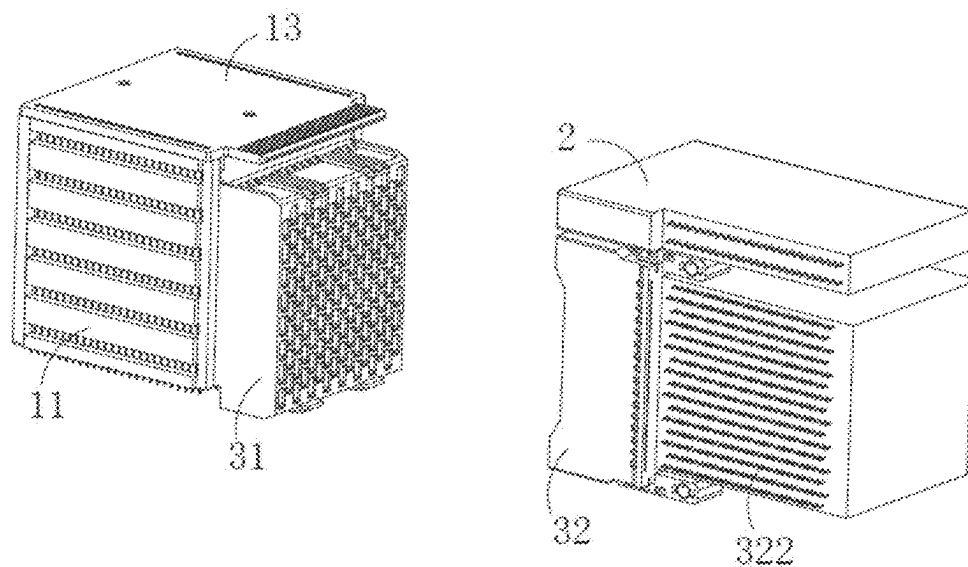


FIG. 5

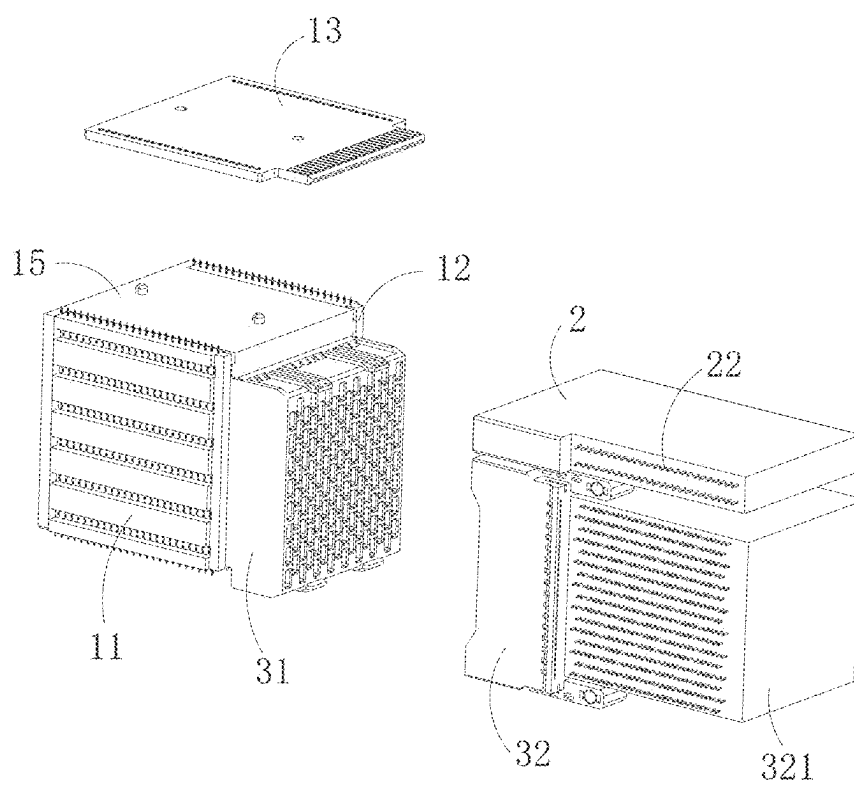


FIG. 6

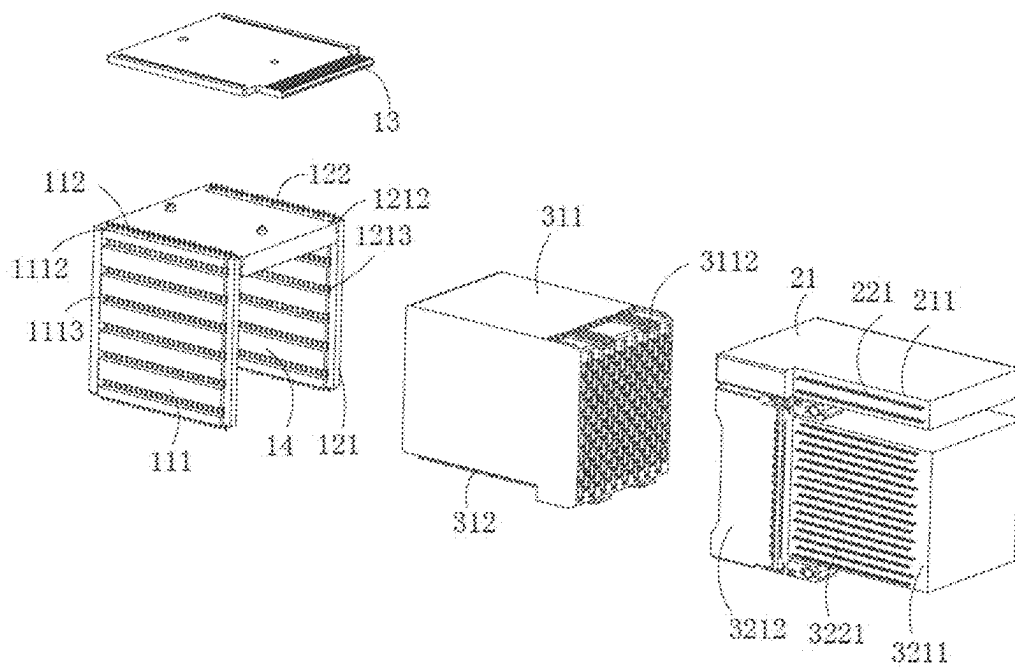


FIG. 7

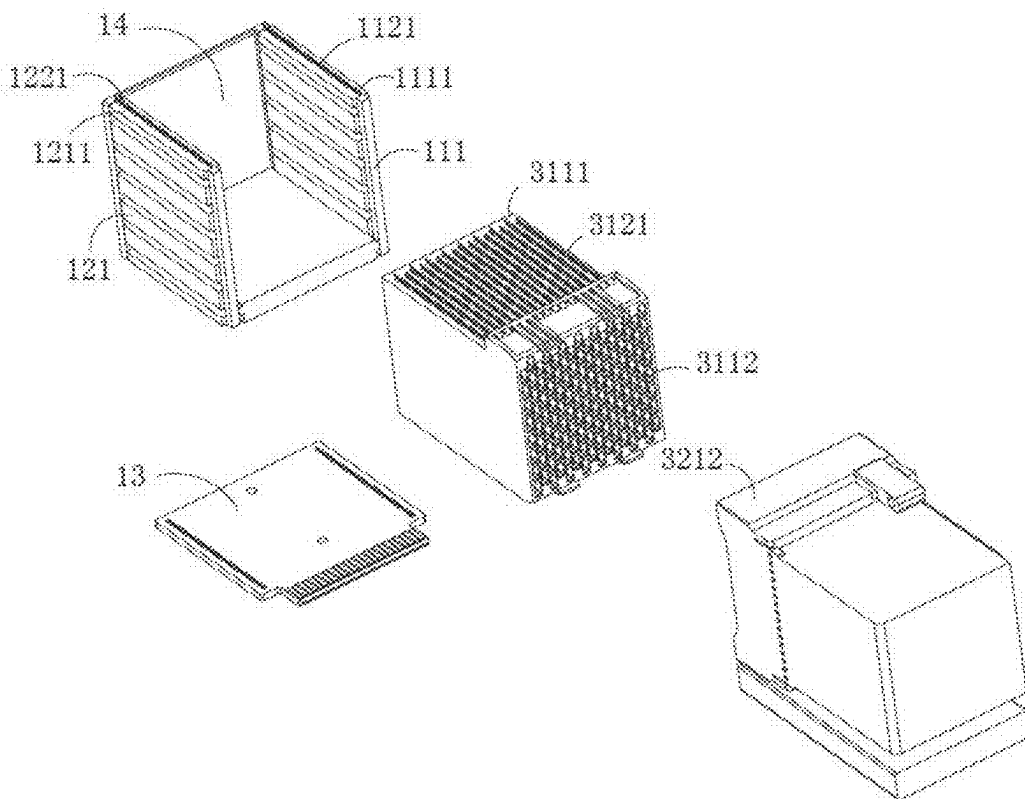


FIG. 8

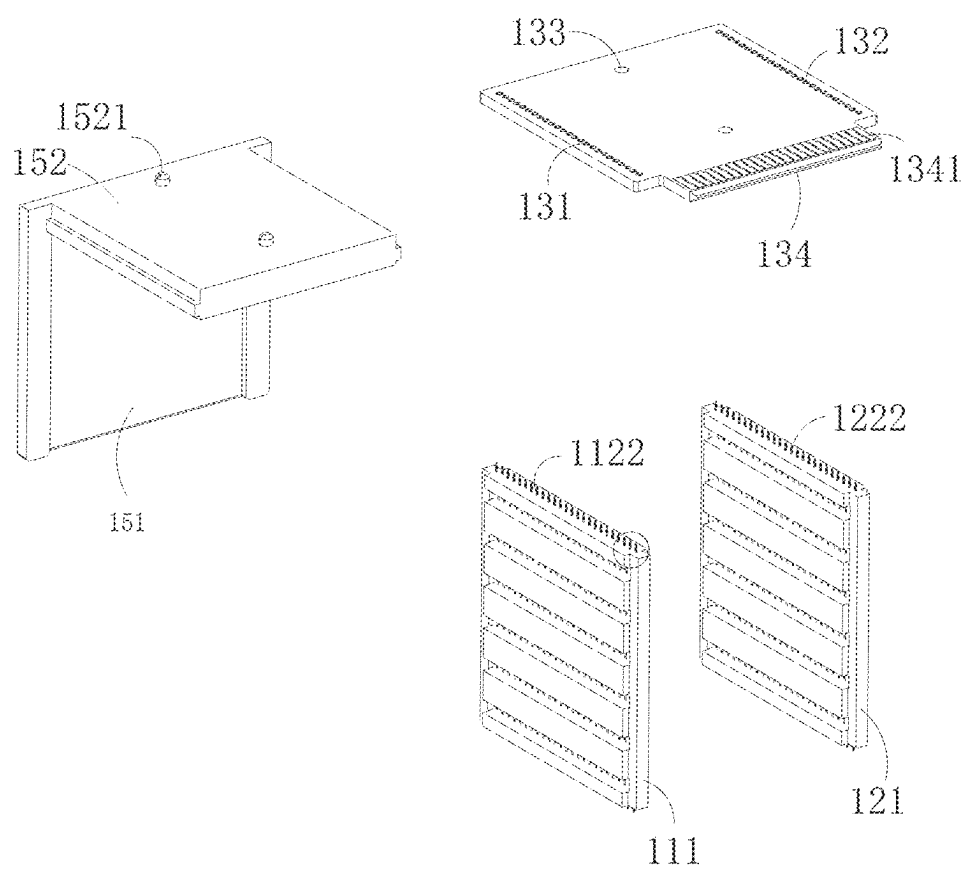


FIG. 9

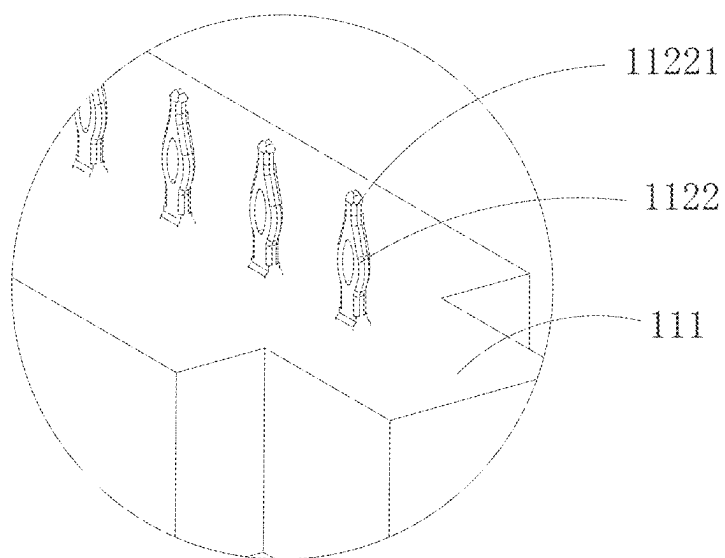


FIG. 10

CONNECTOR ASSEMBLY WITH HIGHER SPACE UTILIZATION

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 17/873,642, filed on Jul. 26, 2022, which claims priority of a Chinese Patent Application No. 202121801823.6, filed on Aug. 3, 2021 and titled “CONNECTOR ASSEMBLY”, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a connector assembly, which belongs to a technical field of connectors.

BACKGROUND

[0003] Some electrical systems often use orthogonal connectors to interconnect two circuit boards with each other. In some applications, the circuit boards are oriented orthogonal to each other. Typically, the orthogonal connector includes a first connector mounted on a first circuit board and a second connector mounted on a second circuit board. The first connector and the second connector are matched and connected to realize the connection between the first circuit board and the second circuit board. The first circuit board and the second circuit board are disposed perpendicular to each other. However, the existing orthogonal connector occupies a relatively large space, resulting in low space utilization in the electrical system.

SUMMARY

[0004] The present disclosure adopts the following technical solution: a connector assembly, including: a card connector including: a first terminal module, the first terminal module including a plurality of first terminals, each first terminal including a first tail pin, the second terminal module being disposed opposite to the first terminal module, the second terminal module including a plurality of second terminals, each second terminal including a second tail pin; and a circuit board card, the circuit board card being electrically connected to the first tail pins of the first terminal module and the second tail pins of the second terminal module; the circuit board including a mating end configured to mate with an interface of a card connector; the mating end including a plurality of conductive elements formed on at least one side of the mating end; wherein the card connector includes a receiving cavity, and the receiving cavity is jointly formed by the first terminal module, the second terminal module and the circuit board card; and a first backplane connector received in the receiving cavity.

[0005] The present disclosure also adopts the following technical solution: a connector assembly, including: a card connector including a first terminal module, a second terminal module parallel to the first terminal module, and a circuit board card disposed on the first terminal module and the second terminal module; the first terminal module and the second terminal module being electrically connected to the circuit board card; the first terminal module, the second terminal module and the circuit board card jointly forming a receiving cavity; and a first backplane connector received in the receiving cavity; wherein the first terminal module includes a first insulating portion and a plurality of first

terminals positioned in the first insulating portion, each first terminal includes a first mounting pin and a first tail pin; the first mounting pin and the first tail pin are disposed oppositely; and wherein the first tail pins and the second tail pins are connected to the circuit board card, the first mounting pins and the second mounting pins are configured to be mounted to a first circuit board.

BRIEF DESCRIPTION OF DRAWINGS

[0006] FIG. 1 is a perspective schematic view of a connector assembly in accordance with an embodiment of the present disclosure;

[0007] FIG. 2 is a perspective schematic view of the connector assembly of FIG. 1 installed on a first circuit board and a second circuit board;

[0008] FIG. 3 is a partially exploded perspective view of FIG. 2;

[0009] FIG. 4 is a partially exploded perspective view of FIG. 1;

[0010] FIG. 5 is another partial perspective exploded view of FIG. 1;

[0011] FIG. 6 is a further partial perspective exploded view of FIG. 5;

[0012] FIG. 7 is a further partial perspective exploded view of FIG. 6;

[0013] FIG. 8 is a partially exploded perspective view of FIG. 7 from another angle;

[0014] FIG. 9 is a partially exploded perspective view of a card connector in FIG. 1; and

[0015] FIG. 10 is an enlarged schematic view of a circled part in FIG. 9.

DETAILED DESCRIPTION

[0016] Exemplary embodiments will be described in detail here, examples of which are shown in drawings. When referring to the drawings below, unless otherwise indicated, same numerals in different drawings represent the same or similar elements. The examples described in the following exemplary embodiments do not represent all embodiments consistent with this application. Rather, they are merely examples of devices and methods consistent with some aspects of the application as detailed in the appended claims.

[0017] The terminology used in this application is only for the purpose of describing particular embodiments, and is not intended to limit this application. The singular forms “a”, “said”, and “the” used in this application and the appended claims are also intended to include plural forms unless the context clearly indicates other meanings.

[0018] It should be understood that the terms “first”, “second” and similar words used in the specification and claims of this application do not represent any order, quantity or importance, but are only used to distinguish different components. Similarly, “an” or “a” and other similar words do not mean a quantity limit, but mean that there is at least one; “multiple” or “a plurality of” means two or more than two. Unless otherwise noted, “front”, “rear”, “lower” and/or “upper” and similar words are for ease of description only and are not limited to one location or one spatial orientation. Similar words such as “include” or “comprise” mean that elements or objects appear before “include” or “comprise” cover elements or objects listed after “include” or “comprise” and their equivalents, and do not exclude other

elements or objects. The term “a plurality of” mentioned in the present disclosure includes two or more.

[0019] Hereinafter, some embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. In the case of no conflict, the following embodiments and features in the embodiments can be combined with each other.

[0020] Referring to FIGS. 1 to 9, the present disclosure discloses a connector assembly 100 including a card connector 1, a plug-in card connector 2 and an orthogonal connector 3. The card connector 1 includes a first terminal module 11, a second terminal module 12 and a circuit board card 13. The circuit board card 13 connects the first terminal module 11 and the second terminal module 12. The circuit board card 13 is lapped with the first terminal module 11 and the second terminal module 12 to jointly form a U-shaped configuration with a receiving cavity 14. An end of the circuit board card 13 is inserted into the plug-in card connector 2. The orthogonal connector 3 is partly received in the receiving cavity 14 and partly arranged in parallel with the plug-in card connector 2, thereby making full use of a peripheral space of the orthogonal connector 3 to improve the space utilization of the connector assembly 100.

[0021] Referring to FIGS. 2 and 3, the connector assembly 100 is mounted on the first circuit board S1 and the second circuit board S2. The orthogonal connector 3 includes a first backplane connector 31 and a second backplane connector 32 for mating with the first backplane connector 31. The card connector 1 and the first backplane connector 31 are mounted on the first circuit board S1. The plug-in card connector 2 and the second backplane connector 32 are mounted on the second circuit board S2. The first backplane connector 31 is received in the receiving cavity 14. In the present embodiment, the first backplane connector 31 and the second backplane connector 32 are mated in an orthogonal manner. The first circuit board S1 is perpendicular to the second circuit board S2. The circuit board card 13 is parallel to the first circuit board S1.

[0022] Referring to FIGS. 1 to 9, the first terminal module 11 and the second terminal module 12 of the card connector 1 are spaced apart and support the circuit board card 13. The circuit board card 13 is erected on the first terminal module 11 and the second terminal module 12. The circuit board card 13 is disposed in parallel with the first circuit board S1. The first terminal module 11 and the second terminal module 12 are symmetrically arranged. The first terminal module 11 includes a plate-shaped first insulating portion 111 and a plurality of first pressing terminals 112 fixed in the first insulating portion 111. In the present embodiment, the first insulating portion 111 is molded together with the first pressing terminals 112 by injection molding. In other embodiments, the first pressing terminals 112 may also be fixed in the first insulating portion 111 in a post-assembly manner. After the card connector 1 is assembled on the first circuit board S1, the first insulating portion 111 is vertically arranged relative to the first circuit board S1. The first insulating portion 111 includes a first bottom end 1111 contacting the first circuit board S1 and a first top end 1112 disposed opposite to the first bottom end 1111. Each first pressing terminal 112 includes a first bottom pressing pin 1121 protruding beyond the first bottom end 1111 and a first top pressing pin 1122 protruding beyond the first top end 1112. In the present embodiment, the first bottom pressing pins 1121 are pressed into corresponding connection holes

(not shown) of the first circuit board S1, and the first pressing pins 1122 are pressed into the first holes 131 of the circuit board card 13. Each of the first bottom pressing pins 1121 and the first top pressing pins 1122 include a fish-eye structure, as shown in FIG. 10. The first pressing pins 1122 are pressed into the first holes 131 of the circuit board card 13 through the fish-eye structures 1121.

[0023] The second terminal module 12 includes a plate-shaped second insulating portion 121 and a plurality of second pressing terminals 122 fixed in the second insulating portion 121. In this embodiment, the second insulating portion 121 is molded together with the second pressing terminals 122 by injection molding. In other embodiments, the second pressing terminal 122 may also be fixed in the second insulating portion 121 in a post-assembly manner. After the card connector 1 is assembled on the first circuit board S1, the second insulating portion 121 is vertically arranged relative to the first circuit board S1. The second insulating portion 121 includes a second bottom end 1211 contacting the first circuit board S1 and a second top end 1212 disposed opposite to the second bottom end 1211. Each second pressing terminal 122 includes a second bottom pressing pin 1221 protruding beyond the second bottom end 1211 and a second top pressing pin 1222 protruding beyond the second top end 1212. In this embodiment, the second bottom pressing pins 1221 are pressed into corresponding connection holes (not shown) of the first circuit board S1. The second top pressing pins 1222 are pressed into the second holes 132 of the circuit board card 13. Each of the second bottom pressing pins 1221 and the second top pressing pins 1222 includes a fish-eye structure. The second bottom pressing pins 1221 and the second top pressing pins 1222 are pressed into the corresponding holes of the second circuit board S2 and the circuit board card 13 through the fish-eye structures. In this embodiment, the first bottom pressing pins 1121 and the second bottom pressing pins 1221 are inserted into the first circuit board S1 by pressing. The electrical connection is achieved by the first bottom pressing pins 1121 and the second bottom pressing pins 1221 elastically contacting the first circuit board S1. In other embodiments, welding can also be used for electrical connection.

[0024] In some embodiments, a plurality of first grooves 1113 are respectively provided on opposite sides of the first insulating portion 111. The first grooves 1113 on the opposite sides are symmetrically arranged. Correspondingly, a plurality of second grooves 1213 are respectively provided on opposite sides of the second insulating portion 121. The second grooves 1213 on the opposite sides are symmetrically arranged. In some embodiments, parts of the first pressing terminals 112 are exposed from the first grooves 1113. Parts of the second pressing terminals 122 are exposed from the second grooves 1213.

[0025] The card connector 1 also includes an inverted L-shaped bracket 15. The bracket 15 includes a rear wall 151 and a top wall 152 extending from an end of the rear wall 151. The rear wall 151 and the top wall 152 are perpendicular to each other. When the card connector 1 is mounted on the first circuit board S1, the rear wall 151 is vertically mounted on the first circuit board S1, and the top wall 152 is suspended in parallel to the first circuit board S1. The first terminal module 11 and the second terminal module 12 are disposed on opposite sides of the bracket 15. The top wall 152 includes at least one positioning post 1521 protruding from a top surface of the top wall 152 for positioning the

circuit board card **13**. In this way, the first terminal module **11**, the second terminal module **12**, the circuit board card **13** and the bracket **15** are enclosed with the first circuit board **S1** to form a receiving cavity **14**. A plurality of first holes **131** and a plurality of second holes **132** are opened on opposite sides of the circuit board card **13**, respectively. The circuit board card **13** defines at least one positioning hole **133** corresponding to the positioning post **1521**, and the positioning post **1521** is inserted into the positioning hole **133**. In this embodiment, the circuit board card **131** defines two positioning holes **133** at intervals. Correspondingly, the top wall **152** includes two protruding positioning posts **1521**. An end of the circuit board card **131** is configured as a plug-in end **134**. Specifically, the circuit board card **131** is installed on the top wall **152** and protrudes beyond the top wall **152** toward an end of an opening of the receiving cavity **14** to be set as a plug-in end **134**. A plurality of conductive pads **1341** are arranged on upper and lower sides of the plug-in end **134**, respectively. The conductive pads **1341** are used to realize electrical connection with the plug-in card connector **2**.

[0026] Referring to FIGS. 1 to 8, the plug-in card connector **2** includes a main body **21** and a plurality of plug-in terminals **22** fixed in the main body **21**. The main body **21** is formed with an insertion slot (not shown in the drawing) for receiving the plug-in end **134** and a mounting end **211** disposed perpendicular to a side where the insertion slot is located. Each plug-in terminal **22** includes a contact portion (not shown in the drawing) extending into the insertion slot and a plug-in pin **221** protruding beyond the mounting end **211**. The plug-in pin **221** is fixed to the second circuit board **S2**.

[0027] Referring to FIGS. 1 to 7, the first backplane connector **31** of the orthogonal connector **3** includes a first housing **311** and a plurality of first backplane terminals **312** assembled in the first housing **311**. The first housing **311** includes a bottom surface **3111** and a first mating end **3112** located outside of the receiving cavity **14**. The bottom surface **3111** is parallel to the first circuit board **S1**. Each first backplane terminal **312** includes a first backplane pin **3121** protruding beyond the bottom surface **3111**. The first backplane connector **31** is partially received in the receiving cavity **14**. The bottom surface **3111** is located in the receiving cavity **14**. The first backplane pins **3121**, the first bottom pressing pins **1121** and the second bottom pressing pins **1221** are arranged in an array and are all mounted to the first circuit board **S1**. The second backplane connector **32** includes a second housing **321** and a plurality of second backplane terminals **322** assembled in the second housing **321**. The second housing **321** includes a side surface **3211** and a second mating end **3212**. Each second backplane terminal **322** includes a second backplane pin **3221** protruding beyond the side surface **3211**. The side surface **3211** is arranged parallel to the second circuit board **S2**. The second backplane pins **3221** and the plug-in pins **221** are arranged in an array and are both mounted to the second circuit board **S2**. The second mating end **3212** and the first mating end **3112** are mated with each other to realize the mating of the first backplane connector **31** and the second backplane connector **32**, and realize the electrical connection of the first circuit board **S1** and the second circuit board **S2**.

[0028] Referring to FIGS. 1 to 9, the card connector **1** is mounted to an edge of the first circuit board **S1**. The plug-in card connector **2** is connected to the circuit board card **13**

and protrudes beyond the first circuit board **S1**. The first backplane connector **31** of the orthogonal connector **3** is installed in the receiving cavity **14** of the card connector **1** and located at the edge of the first circuit board **S1**. The second backplane connection **32** protrudes beyond the first circuit board **S1**. Both the card connector **1** and the first backplane connector **31** are mounted to the first circuit board **S1**. The card connector **1** is arranged around the first backplane connector **31**. Both the plug-in card connector **2** and the second backplane connector **32** are mounted to the second circuit board **S2**. The plug-in card connector **2** and the second backplane connector **32** are stacked relative to the first circuit board **S1** and are arranged side by side on the second circuit board **S2**. In this way, the space can be fully utilized and the space utilization can be improved.

[0029] The above embodiments are only used to illustrate the present disclosure and not to limit the technical solutions described in the present disclosure. The understanding of this specification should be based on those skilled in the art. Descriptions of directions, although they have been described in detail in the above-mentioned embodiments of the present disclosure, those skilled in the art should understand that modifications or equivalent substitutions can still be made to the application, and all technical solutions and improvements that do not depart from the spirit and scope of the application should be covered by the claims of the application.

What is claimed is:

1. A connector assembly, comprising:

a card connector comprising:

- a first terminal module, the first terminal module comprising a plurality of first terminals, each first terminal comprising a first tail pin;
- a second terminal module, the second terminal module being disposed opposite to the first terminal module, the second terminal module comprising a plurality of second terminals, each second terminal comprising a second tail pin; and
- a circuit board card, the circuit board card being electrically connected to the first tail pins of the first terminal module and the second tail pins of the second terminal module; the circuit board comprising a mating end configured to be mated with an interface of a card connector; the mating end comprising a plurality of conductive elements formed on at least one side of the mating end;

wherein the card connector comprises a receiving cavity, and the receiving cavity is jointly formed by the first terminal module, the second terminal module and the circuit board card; and

a first backplane connector received in the receiving cavity.

2. The connector assembly according to claim 1, wherein the first backplane connector comprises a first housing and a plurality of first backplane terminals at least partially received in the first housing; each first backplane terminal comprises a first backplane pin; the first backplane pin extends along a direction opposite to the first tail pin and the second tail pin.

3. The connector assembly according to claim 2, wherein each first terminal comprises a first pressing pin; the first pressing pins and the first tail pins are disposed on opposite sides of the first terminal module, respectively;

each second terminal comprises a second pressing pin; the second pressing pins and the second tail pins are disposed on opposite sides of the second terminal module, respectively;

wherein the first pressing pins, the second pressing pins and the first backplane pins are configured to be mounted to a first circuit board.

4. The connector assembly according to claim 3, wherein the first terminal module further comprises a first insulating portion, the first terminals are disposed on the first insulating portion, the first tail pins and the first pressing pins extend beyond the first insulating portion;

the second terminal module comprises a second insulating portion, the second terminals are disposed on the second insulating portion, the second tail pins and the second pressing pins extend beyond the second insulating portion.

5. The connector assembly according to claim 1, further comprising a bracket, the bracket comprising a first wall disposed between the first terminal module and the second terminal module; the first wall comprising at least one positioning post;

the circuit board card defining at least one positioning hole to receive the at least one positioning post.

6. The connector assembly according to claim 5, wherein the first wall, the first terminal module and the second terminal module jointly form a U-shaped configuration.

7. The connector assembly according to claim 5, wherein the bracket is of an L-shaped configuration, and comprises a second wall that is perpendicular to the first wall; the second wall is disposed at a rear side of the receiving cavity.

8. The connector assembly according to claim 1, wherein the circuit board card defines a plurality of first conductive holes and a plurality of second conductive holes;

the plurality of first conductive holes are disposed in a first line; the first tail pins are pressed into the plurality of first conductive holes;

the plurality of second conductive holes are disposed in a second line; the second tail pins are pressed into the plurality of second conductive holes.

9. The connector assembly according to claim 1, further comprising the card connector and a second backplane connector, wherein the card connector is a plug-in card connector, the second backplane connector and the card connector are disposed in parallel, the first backplane connector and the second backplane connector are mated in an orthogonal manner.

10. The connector assembly according to claim 9, wherein the connector assembly is configured to be mounted on a first circuit board and a second circuit board, the card connector and the first backplane connector are mounted on the first circuit board, the card connector and the second backplane connector are mounted on the second circuit board, and the first circuit board is perpendicular to the second circuit board.

11. The connector assembly according to claim 1, wherein the first terminal module and the second terminal module are spaced apart and support the circuit board card; and wherein the circuit board card is erected on the first terminal module and the second terminal module.

12. A connector assembly, comprising:

a card connector comprising a first terminal module, a second terminal module parallel to the first terminal module, and a circuit board card disposed on the first

terminal module and the second terminal module; the first terminal module and the second terminal module being electrically connected to the circuit board card; the first terminal module, the second terminal module and the circuit board card jointly forming a receiving cavity; and

a first backplane connector received in the receiving cavity;

wherein the first terminal module comprises a first insulating portion and a plurality of first terminals positioned in the first insulating portion, each first terminal comprises a first mounting pin and a first tail pin; the first mounting pin and the first tail pin are disposed oppositely;

wherein the second terminal module comprises a second insulating portion and a plurality of second terminals positioned in the second insulating portion, each second terminal comprises a second pressing pin and a second tail pin; the second mounting pin and the second tail pin are disposed oppositely; and

wherein the first tail pins and the second tail pins are connected to the circuit board card, the first mounting pins and the second mounting pins are configured to be mounted to a first circuit board.

13. The connector assembly according to claim 12, wherein the circuit board card defines a plurality of first holes and a plurality of second holes; and the first tail pins are pressed into the first holes, and the second tail pins are pressed into the second holes.

14. The connector assembly according to claim 12, further comprising a plug-in card connector and an orthogonal connector, the orthogonal connector comprising the first backplane connector and a second backplane connector, an end of the circuit board card being inserted into the plug-in card connector, the first backplane connector and the second backplane connector being mated in an orthogonal manner.

15. The connector assembly according to claim 14, further comprising the first circuit board and a second circuit board, the card connector and the first backplane connector being mounted on the first circuit board, the plug-in card connector and the second backplane connector being mounted on the second circuit board, the first circuit board being perpendicular to the second circuit board.

16. The connector assembly according to claim 14, wherein the first backplane connector comprises a first housing and a plurality of first backplane terminals assembled in the first housing; wherein the second backplane connector comprises a second housing and a plurality of second backplane terminals assembled in the second housing; and wherein the first backplane terminals are configured to be mounted to the first circuit board, and the second backplane terminals are configured to be mounted to the second circuit board.

17. The connector assembly according to claim 12, wherein the first terminal module and the second terminal module are spaced apart and support the circuit board card.

18. The connector assembly according to claim 12, wherein the card connector further comprises a bracket, the bracket comprises a top wall, the first terminal module and the second terminal module are disposed on opposite sides of the bracket, and the circuit board card is positioned on the top wall.

19. The connector assembly according to claim **18**, wherein the circuit board card defines a positioning hole, and the top wall comprises a positioning post inserted in the positioning hole.

20. The connector assembly according to claim **18**, wherein an end of the circuit board card protruding beyond the top wall is set as a plug-in end, a plurality of conductive elements are disposed on the plug-in end, and the conductive elements are configured to be electrically connected with a plug-in card connector.

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